

# Applied Statistics

Probability interpretation, point estimates

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Laboratory for method of big data analysis (Lambda)



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# In this lecture

- ▶ Interpretations of Probability: Bayesian and Frequentist Inference
- ▶ Point Estimates:
  - method of moments;
  - maximum likelihood method;
  - maximum a posteriori method.

# Probability



# What is probability

*Probability is the most important concept in modern science, especially as nobody has the slightest notion what it means.*

*—Bertrand Russell, 1929 Lecture*

# Mathematical Approach

- ▶ Kolmogorov axioms:

- $\mathcal{P}(A) > 0$ ;
- $\mathcal{P}(\text{True}) = 1$ ;
- $\mathcal{P}(A \cup B) = \mathcal{P}(A) + \mathcal{P}(B)$  (if  $A$  and  $B$  do not cross).

- ▶ Bayes Theorem:

- $\mathcal{P}(A|B) = \frac{\mathcal{P}(B|A)\mathcal{P}(A)}{\mathcal{P}(B)}$ , где  $\mathcal{P}(A|B)$  - probability of A given that B is fulfilled.

- ▶ Full probability:

- $\mathcal{P}(A) = \sum_n \mathcal{P}(A \cap B_n) \mathcal{P}(B_n)$ , where  $B_n$  partitioning of the entire space.

# Outcomes

- ▶ Probability is a concept that exists without reference to mathematical objects.
- ▶ In applied statistics, a mathematized version of the concept is used, often reduced to Kolmogorov's axioms.
- ▶ Interpreting the resulting mathematical expressions from a real-world perspective is a matter of additional reasoning.

E. T. Jaynes, Probability Theory: The Logic of Science

# Interpretation of probability



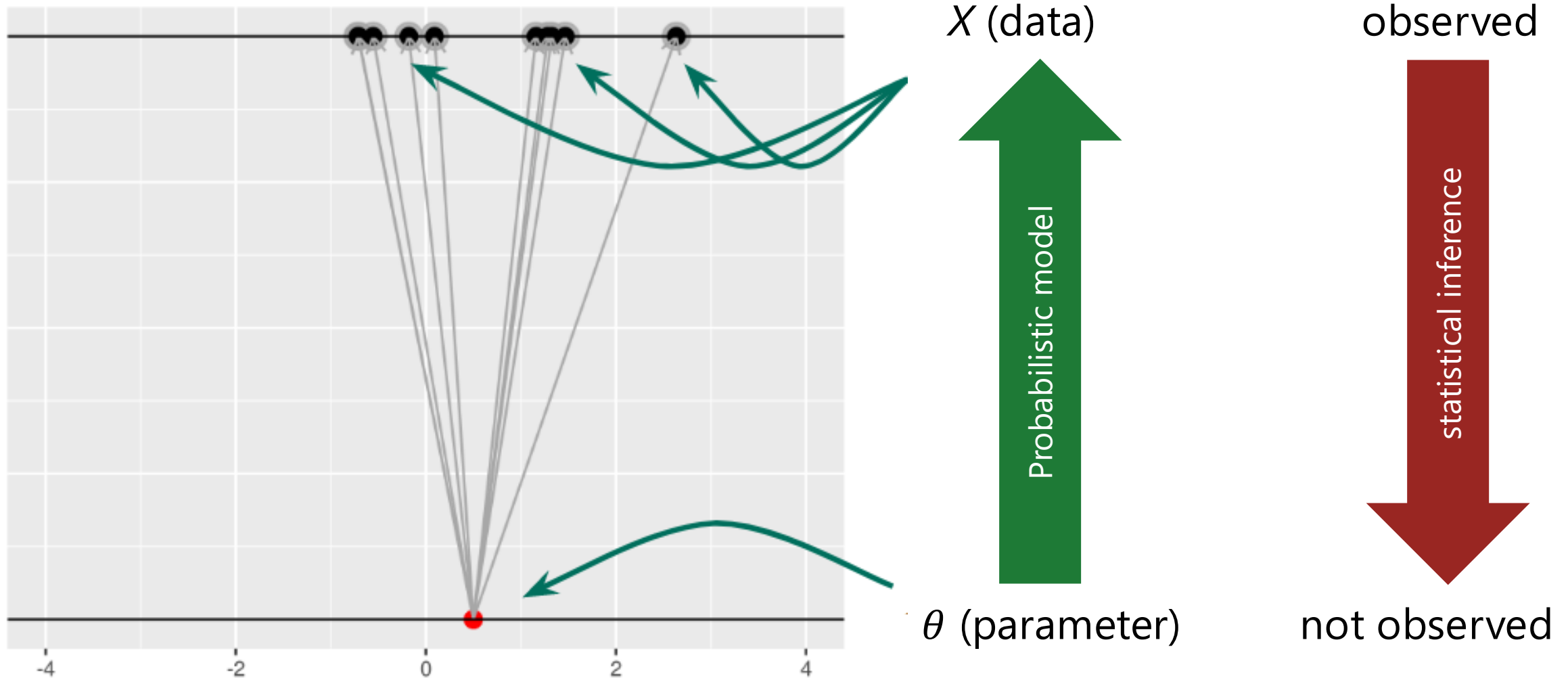
# Why interpret numbers?

- ▶ Applied statistics combines mathematical theory and real-world problems.
- ▶ For real life, it's not just the numerical value that's important, but also the understanding of how that value answers the question.
- ▶ The question of interpretation is close to philosophy but has a direct bearing on the results (as we'll see later).

The Oxford Handbook of Probability and Philosophy



# Statement of the (inverse) problem



# Frequentist interpretation

Classical (частотная, frequentist):

$$\mathcal{P}(X) = \lim_{N \rightarrow \infty} \frac{n}{N}$$

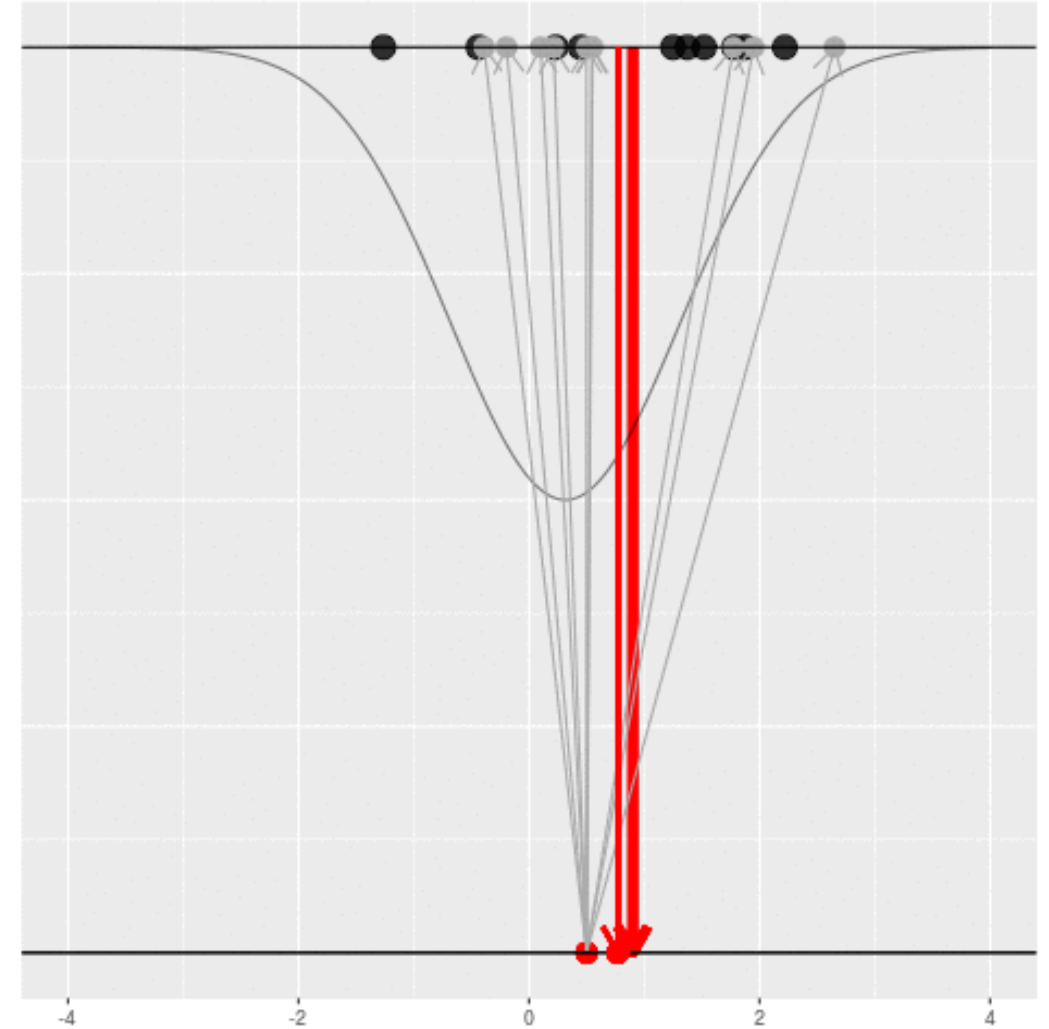
where  $n$  – number of events that occurred  $X$  out of  $N$  tries.

- ▶ Randomness is objective.
- ▶ The conclusion is based on a large number of experiments.
- ▶ The parameters are fixed.

# Frequentist interpretation: consequences

- ▶ The parameters are fixed.
- ▶ The inference is based on the idea of randomness of the data.
- ▶ Mathematical result:

$$\mathcal{P}(X|\theta)$$



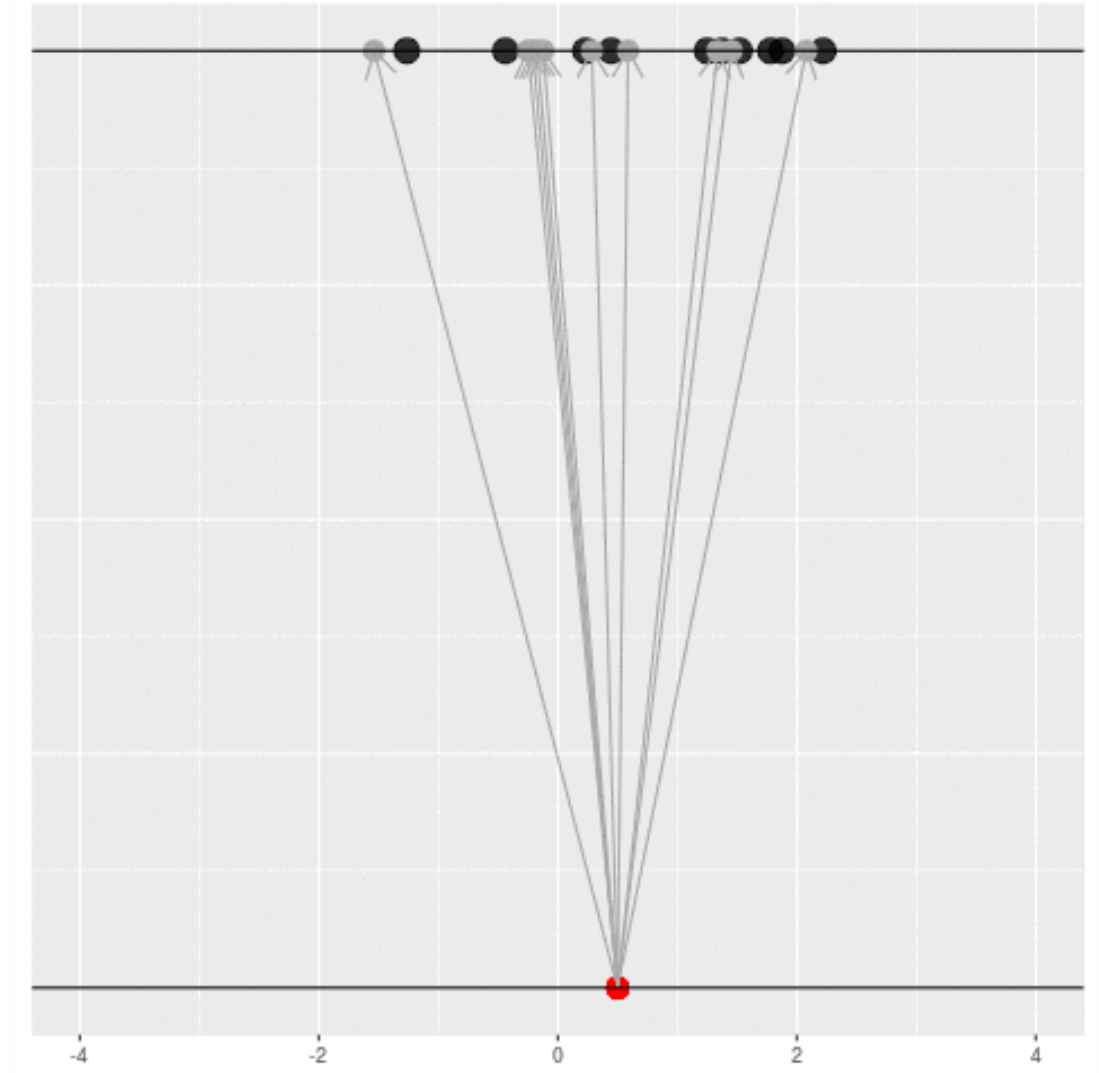
# Bayesian interpretation

$\mathcal{P}(X)$  – degree of confidence in the outcome  $X$ .

- ▶ Randomness is the measure of our ignorance.
- ▶ The conclusion is based on a single experiment.
- ▶ The parameters are uncertain, meaning they are random from our perspective.

# Bayesian interpretation: consequences

- ▶ The parameters are random (from our point of view).
- ▶ The conclusion is based on the idea of true data.

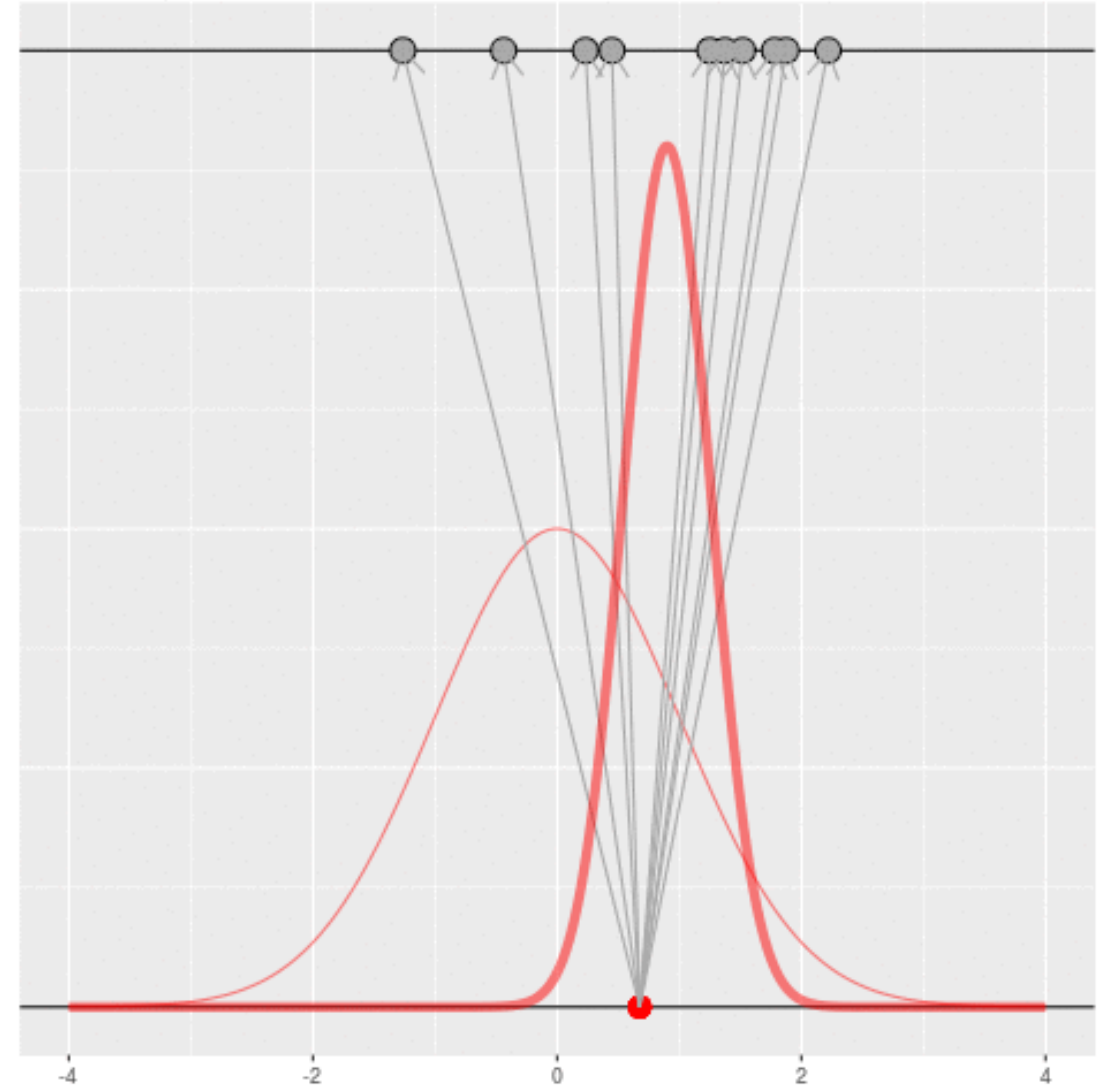


# Bayesian interpretation: consequences

- ▶ **A priori** knowledge of the distribution of parameters  $\mathcal{P}(\theta)$ .
- ▶ For each  $\theta$  exists data distribution  $\mathcal{P}(X|\theta)$ .
- ▶ **Use** Bayes theorem:

$$\mathcal{P}(\theta|X) = \frac{\mathcal{P}(X|\theta)\mathcal{P}(\theta)}{\mathcal{P}(X)}$$

- ▶ Obtain **a posteriori** parameter distribution  $\theta$ .

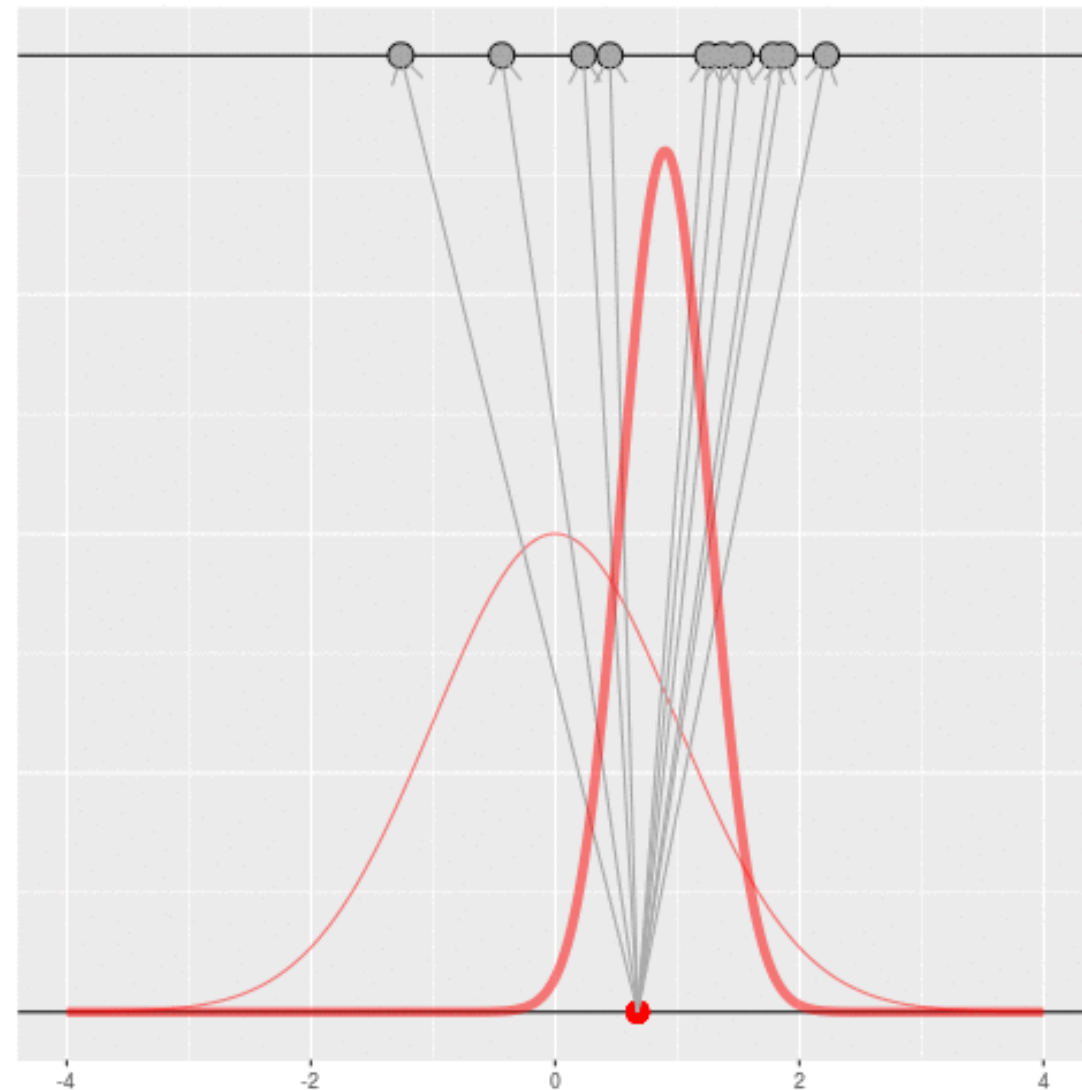


# Bayesian interpretation: consequences

**A priori**

Measurement

**A posteriori**



Ryan Giordano's blog

# Comparison of interpretations

## Classics

- ▶ Arbitrariness in the choice of method.
- ▶ Uses asymptotic characteristics.
- ▶ Satisfies Kolmogorov's axioms.
- ▶ Method response:  $\mathcal{P}(X|\theta)$ .

## Bayes

- ▶ Arbitrariness in the choice of prior probability.
- ▶ Uses Bayes' theorem.
- ▶ Satisfies Kolmogorov's axioms.
- ▶ Method response:  $\mathcal{P}(\theta|X)$ .

**What question do we usually ask?**

B. Efron, T. Hastie, Computer Age Statistical Inference



# Coin toss

**Given:** 14 coin tosses, heads came up 10 times.

**Find:** Probability of two consecutive heads.

# Coin Toss: Frequency Solution

**Given:** 14-coin tosses, heads came up 10 times.

**Find:** Probability of two consecutive heads.

Let's estimate the probability of getting heads:

$$\hat{p} = \frac{10}{14} \approx 0.71$$

Then the probability of two heads in a row is:

$$\hat{p}^2 \approx 51\%$$

# Броски монетки: байесовское решение (1/2)

**Given:** 14-coin tosses, heads came up 10 times.

**Find:** Probability of two consecutive heads.

Let us estimate the probability of obtaining data for some  $p$ :

$$\mathcal{P}(X|p) = \binom{14}{10} p^{10} (1-p)^4$$

The prior probability for  $p$ :

$$p \sim \text{Uniform}(0; 1) = \text{Beta}(p, 1, 1)$$

The probability  $\mathcal{P}(X)$  does not depend on  $p$  and (for us) is constant.

Then:

$$\mathcal{P}(p|X) = \frac{\mathcal{P}(X|p)\mathcal{P}(p)}{\mathcal{P}(X)} \sim p^{10}(1-p)^4$$

# Броски монетки: байесовское решение (2/2)

**Given:** 14-coin tosses, heads came up 10 times.

**Find:** Probability of two consecutive heads.

Let's find the answer:

$$\mathcal{P}(2 \text{ орла} | X) = \int P(2 \text{ орла} | p) P(p | X) dp = \int p^{12} (1 - p)^4 dp \approx 49\%$$

# Броски монетки: обсуждение

**Дано:** 14 бросков монет, 10 раз выпал орёл.

**Найти:** Вероятность двух следующих орлов подряд.

- ▶ Оба решения правильные.
- ▶ Заменой **априорного** распределения в байесовском методе можно добиться совпадения ответов.
- ▶ Интерпретация вероятностей имеет решающее влияние на ответ.

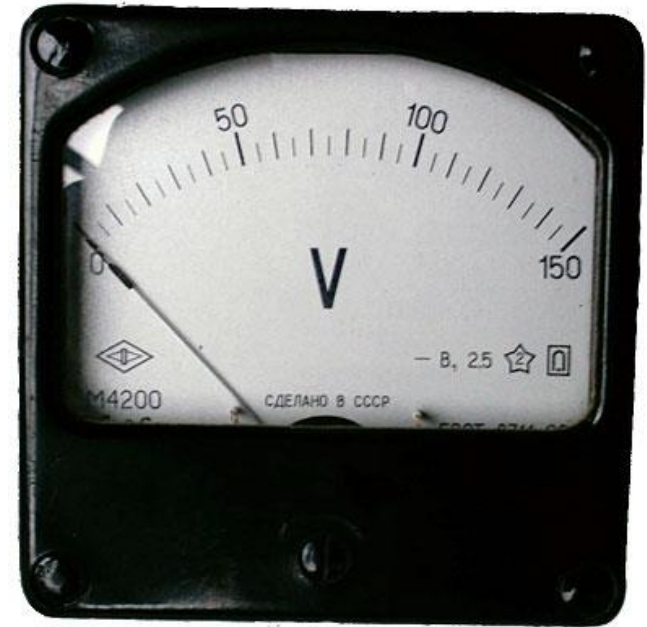
<https://www.behind-the-enemy-lines.com/2008/01/are-you-bayesian-or-frequentist-or.html>

# Voltmeter readings

- ▶ The engineer takes readings from several samples:

$$x_i \sim N(\mu, 4).$$

- ▶ Measurements  $82 < x_i < 99$  B, unbiased estimate  $\mu$   
sample mean  $\bar{x}=92$  B.
- ▶ It turns out that the voltmeter can't take readings  
anymore 100 B (arrow sticks).
- ▶ **Frequentist approach** says the analysis needs to be  
repeated taking into account new information.
- ▶ **Bayesian approach подход** uses a new prior  
distribution.



# Outcome

- ▶ The main interpretations of the probabilistic picture lead to fundamentally different answers.
- ▶ No one interpretation is the only correct one.
- ▶ The use of certain methods depends on the question posed.

Deborah Mayo, Statistical Inference as Severe Testing (How to Get Beyond the Statistics Wars)

# Point Estimate





# Problem Statement

- ▶ **Parametric estimation problem**: it is necessary to estimate the value of  $T(\theta)$ , where  $T$  is some function of the parameter  $\theta$ :

$$T: \Theta \rightarrow \mathcal{Y}$$

$$\theta \mapsto T(\theta)$$

it is necessary to estimate the value of  $T(\theta)$ , where  $T$  is some function of the parameter

$$\hat{T}: \mathcal{X} \rightarrow \hat{\mathcal{Y}}.$$

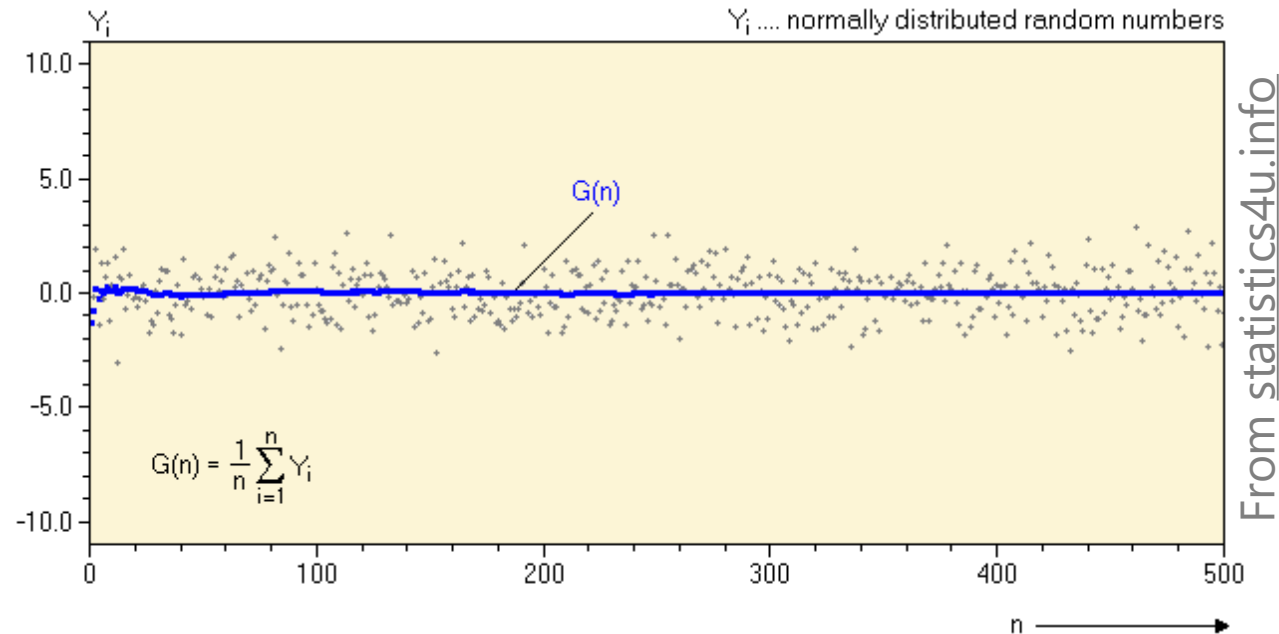
- ▶ The estimate should be:
  - Consistent  $\hat{T}_n \rightarrow T$ .
  - Unbiased  $\mathbb{E}\hat{T}_n - T = 0$ .
  - Efficient  $\text{Var}\hat{T}_n \rightarrow \min$ .

# Sample mean

For iid  $x_i \sim f$  quantities, we can plot the sample mean

$$\bar{x} = \frac{1}{n} \sum x_i.$$

Assuming  $f \sim N(\mu, \sigma)$ ,  $\bar{x}$  is a good estimate for  $\mu$ .



# Cauchy distribution

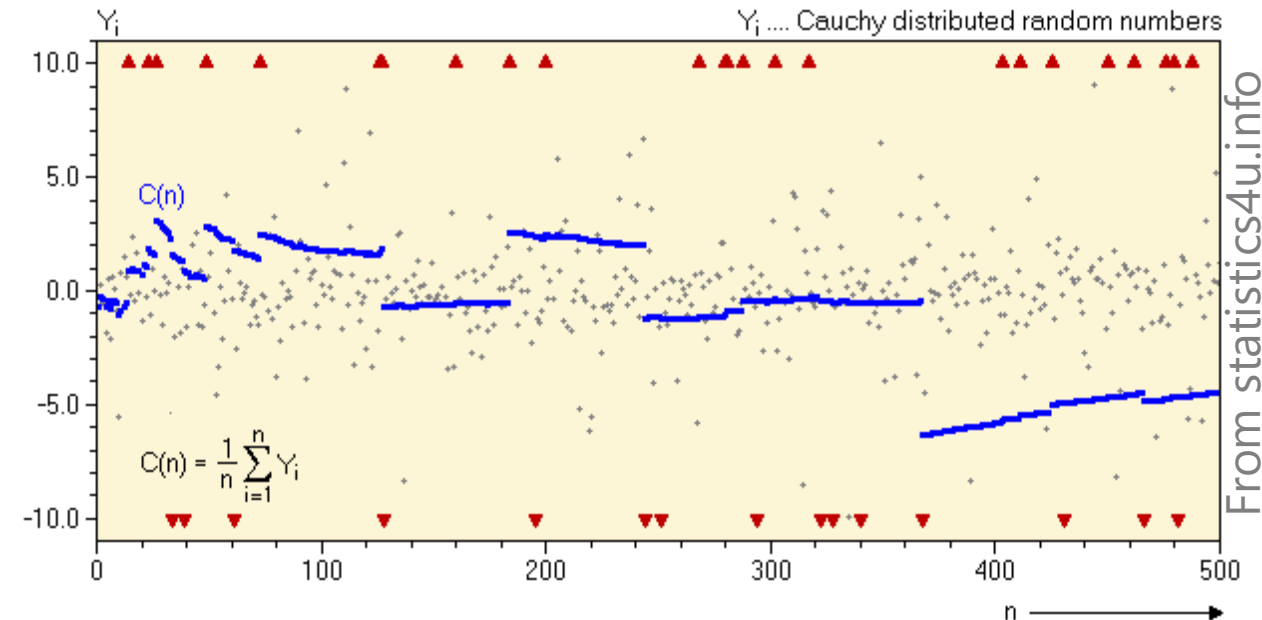
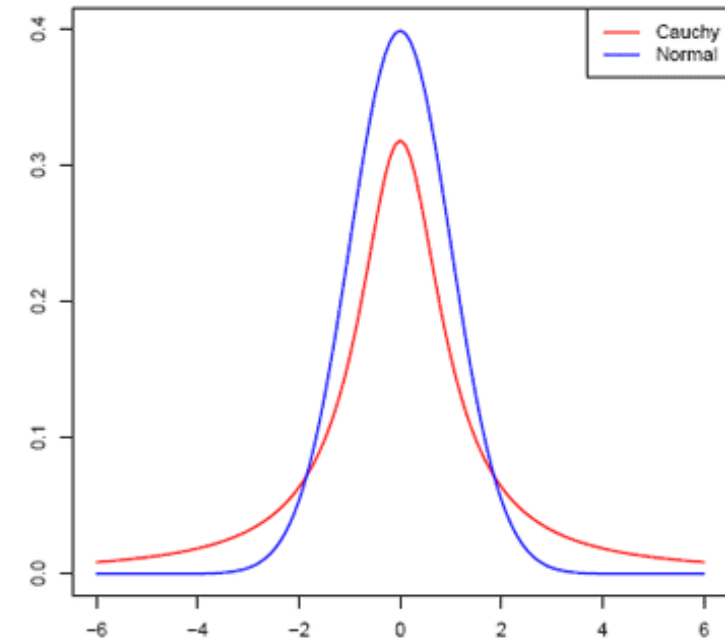
$$f(x; x_0, \gamma) = \frac{1}{\pi} \left( \frac{\gamma}{(x - x_0)^2 + \gamma^2} \right)$$

$\gamma$  – scale parameter;  
 $x_0$  – shift parameter.

The shift parameter is estimated  
by a measure of central  
tendency.

But:

$$\mathbb{E}x = \infty$$



# Stable central tendencies

- ▶ **Sample median;**
- ▶  **$\alpha$ -trimmed sample mean:**
  - Remove  $\alpha$  extreme values  $\alpha$ -
- ▶ **winsorized sample mean**

Replace  $\alpha$  of extreme values with the last non-extremal ones.

- ▶ **Hodges-Lehman estimate**
  - Sample median :  $\{\frac{x_i+x_j}{2}; 1 \leq i \leq j \leq n\}$

# Method of moments

Пусть  $\theta$  – параметр распределения  $F_\theta(x)$ , определим  **$j$  момент**:

$$\alpha_j = \mathbb{E}(X^j) = \int x^j dF_\theta(x),$$

и  **$j$ -й выборочный момент** выборки  $x_1, \dots, x_n \sim F_\theta$  согласно формуле:

$$\hat{\alpha}_j = \frac{1}{n} \sum x_i^j.$$

Тогда

$$\alpha_j(\hat{\theta}) = \hat{\alpha}_j$$

Позволяет получить несмещённую асимптотически нормальную оценку.

# Example: Method of moments

Пусть  $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ , тогда

$$\alpha_1 = \mathbb{E}(X_1) = \mu,$$

$$\alpha_2 = \mathbb{E}(X_1^2) = \text{Var}(X_1) + (\mathbb{E}(X_1))^2 = \sigma^2 + \mu^2,$$

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i,$$

$$\hat{\sigma}^2 + \hat{\mu}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2.$$

Решая систему уравнений, получаем, что

$$\hat{\mu} = \overline{X}_n \quad \text{и} \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \overline{X}_n)^2.$$

# Outcome

- ▶ When analyzing real data, the robustness of the estimate plays an important role.
- ▶ To construct robust estimates, you can use one of several approaches known in the literature \*.

# Maximum likelihood estimate





# Maximum likelihood estimation

Let there be iid sample  $x_1, \dots, x_n \sim f_\theta$ ,  $\theta$  – distribution density parameter  $f_\theta$ . We will call the likelihood function:

$$\mathcal{L}_n(\theta) = \prod_{i=1}^n f_\theta(x_i).$$

The maximum likelihood estimate (MLE) is defined as that value  $\hat{\theta}_n$  of the parameter  $\theta$  that maximizes  $\mathcal{L}_n(\theta)$ .

# Maximum likelihood estimation

ОМП состоятельная, то есть  $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_*$ , где  $\theta_*$  — реальное значение параметра  $\theta$ ;

ОМП не зависит от параметризации, то есть  $\hat{\theta}_n$  — ОМП для  $\theta$ , тогда  $g(\hat{\theta}_n)$  — ОМП для  $g(\theta)$ ;

ОМП асимптотически нормальна:  $(\hat{\theta} - \theta_*)/\hat{se} \rightsquigarrow \mathcal{N}(0, 1)$ ;

ОМП асимптотически оптимальна или эффективна (при достаточно большом объеме выборки ОМП имеет меньшую дисперсию).

# Weibull distribution likelihood

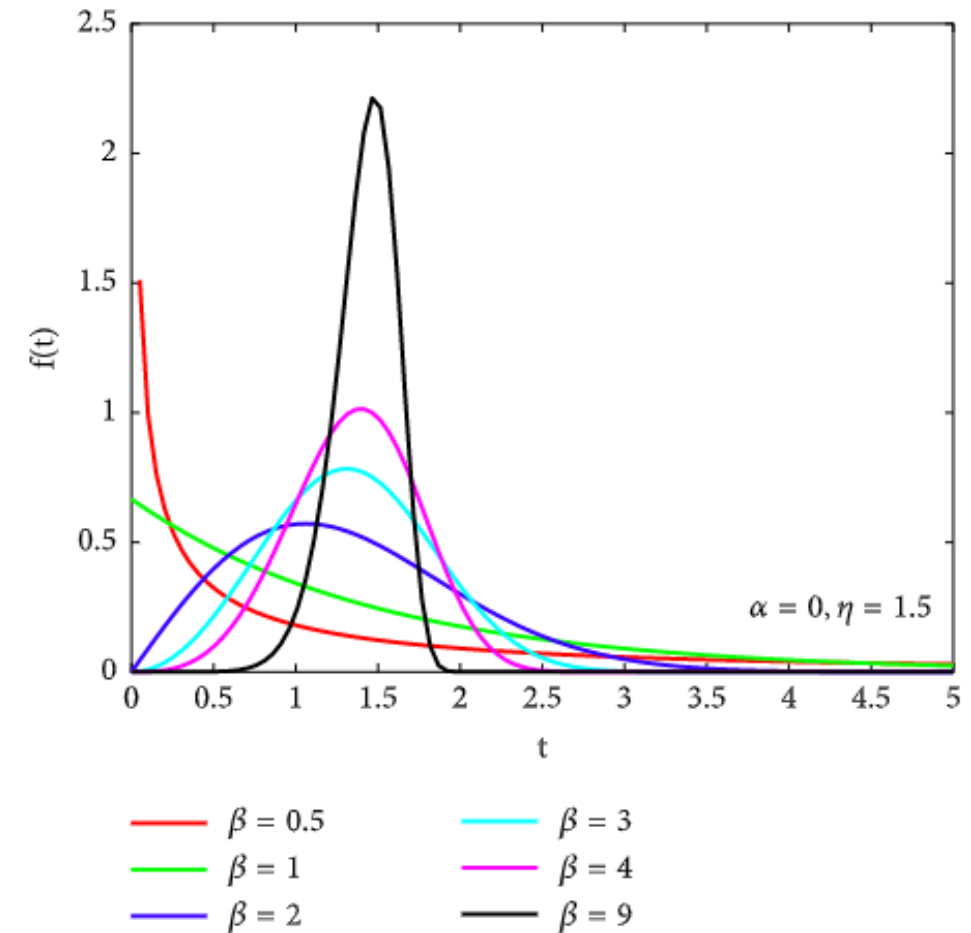
$$F(t, \alpha, \beta, \eta) = \begin{cases} 1 - e^{-((t-\alpha)/\eta)^\beta}, & t > \alpha \\ 0, & t \leq \alpha \end{cases}$$

Let's find the maximum:

$$\sum_{i=1}^n \left[ \frac{1}{\beta} + \ln(t_i - \alpha) - \ln(\eta) - \left( \frac{(t_i - \alpha)}{\eta} \right)^\beta \ln \left( \frac{(t_i - \alpha)}{\eta} \right) \right] = 0$$

$$\sum_{i=1}^n \left[ -\frac{\beta}{\eta} + \left( \frac{\beta}{\eta} \right) \left( \frac{(t_i - \alpha)}{\eta} \right)^\beta \right] = 0$$

$$\sum_{i=1}^n \left[ -\frac{(\beta - 1)}{(t_i - \alpha)} + \left( \frac{\beta}{\eta} \right) \left( \frac{(t_i - \alpha)}{\eta} \right)^{\beta-1} \right] = 0$$



It is not possible to find solutions to this system of equations.

# Outcome

- ▶ Maximum likelihood estimation has many useful theoretical properties.
- ▶ In practice, even fairly simple functions can lead to biased estimates.
- ▶ There is a dependence on model assumptions.

# Maximum a posteriori estimate



# Construction of the assessment

- ▶ MLE determines the parameter values at which our data are most likely.
- ▶ In fact, we usually ask ourselves which parameter values are most likely:

$$f(\theta|X) = f(X|\theta)g(\theta)/h(X) ,$$

where  $f$ ,  $g$ , and  $h$  – are the corresponding distribution functions.

- ▶ **Estimation of the posterior maximum (MAP)** is defined as the value  $\hat{\theta}_n$  of parameter  $\theta$  that maximizes  $f(\theta|X)$ .

# Connection to MLE

- ▶ MLE and MAP are connected:

$$f(\theta|X) = \frac{f(X|\theta)g(\theta)}{h(X)} = \prod \frac{f(x_i; \theta)g(\theta)}{h(x)} \sim c \prod f(x_i; \theta)g(\theta)$$

- ▶ Take log:

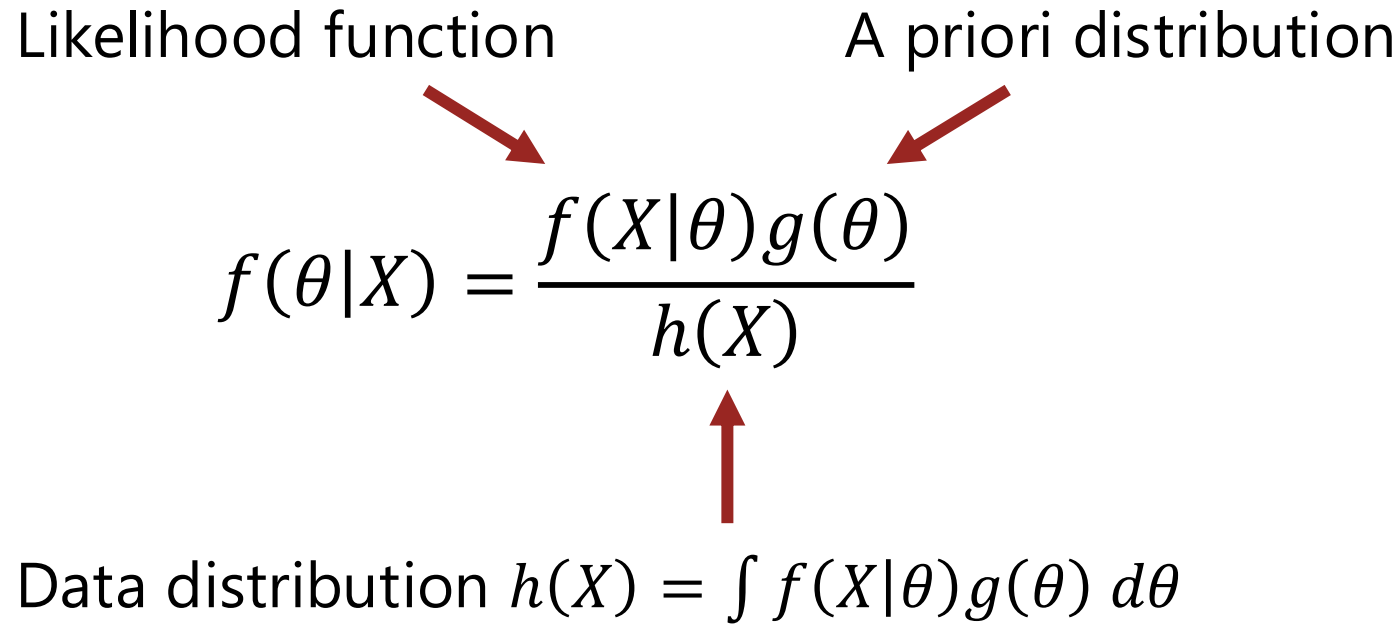
$$\log f(\theta|X) = \log(g(\theta)) + \sum \log(f(x_i|\theta)).$$

- ▶ Means MAP estimate and MLE coincide up to an a priori estimate  $\log(g(\theta))$ .
- ▶ How to choose  $g(\theta)$ ?

# Conjugate prior distributions

Likelihood function

A priori distribution


$$f(\theta|X) = \frac{f(X|\theta)g(\theta)}{h(X)}$$

Data distribution  $h(X) = \int f(X|\theta)g(\theta) d\theta$

It is possible to select a priori distribution  $g(\theta)$  that does not output when multiplied by  $f(x|\theta)$ , leaving  $f(\theta | X)$  in the same class of distributions.

We will call the family  $g(\theta)$  **conjugate** to  $f(x|\theta)$ .



# Coin toss

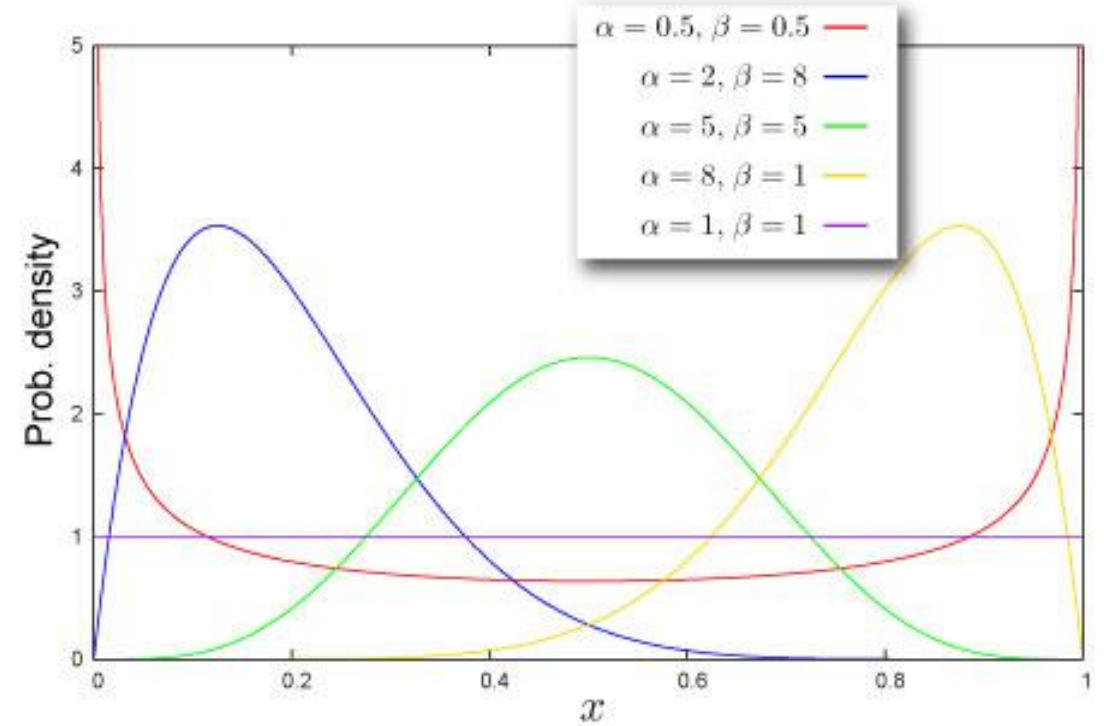
$$\mathcal{P}(p|X) = \frac{\mathcal{P}(X|p)\mathcal{P}(p)}{\mathcal{P}(X)}$$

Для  $s$  орлов и  $f$  решек

$$\mathcal{P}(X|p) = \binom{s+f}{f} p^s (1-p)^f$$

$$\mathcal{P}(p) = \text{Uniform}(0; 1) = \frac{1}{B(\alpha, \beta)} p^{\alpha-1} (1-p)^{\beta-1} \equiv \text{Beta}(p, \alpha, \beta)$$

$$B(\alpha, \beta) = \int p^{\alpha-1} (1-p)^{\beta-1}$$



# Coin Toss

$$\mathcal{P}(p|X) = \frac{\mathcal{P}(X|p)\mathcal{P}(p)}{\mathcal{P}(X)}$$

Multiplying and normalizing, we get

$$\mathcal{P}(p|X) = \frac{1}{B(s+\alpha, f+\beta)} p^{s+\alpha-1} (1-p)^{f+\beta-1} \equiv \text{Beta}(p, \alpha + s, \beta + f)$$

Beta is the **conjugate prior distribution** for the Bernoulli distribution.

# Outcome

- ▶ MAP allows one to take into account previous measurements.
- ▶ The MAP result depends on the parameterization.
- ▶ Selecting a priori distribution can significantly speed up the solution of the problem\*.

# General outcome

- ▶ Probabilities can be interpreted in different ways. The leading interpretations are Classical (Frequentist) and Bayesian.
- ▶ There are several methods for point parametric estimation: the method of moments, the maximum likelihood method, and the a posteriori maximum method are leading
- ▶ Each method has problems when applied to real data.